Hello friends,

I hope you're all doing well.
The following exercises are not exam questions, but rather open-ended prompts whose answers can often be found online, through chatbots, or even in technical interviews. These are intended to encourage deeper exploration and self-directed learning based on your current knowledge.

We've selected these questions based on real-world experience and past exam patterns to help you better understand the types of concepts that appear in the LPIC exam. Many of the questions don't have a single correct answer — instead, they are designed to broaden your perspective and understanding.

Please be kind and patient with yourself as you write and research your responses. Please complete the following exercises using the same format as the directory structure below. Combine your answers into a single PDF file named answer.pdf **and submit a pull request.**

📁 Reference format:
https://github.com/devopsdoctors/Academy/tree/main/L1-JavanPahlevanan/Exercises/T(x)/name-family(emailAddress)/answer.pdf

📄 Sample answer file:
https://github.com/devopsdoctors/Academy/tree/main/L1-JavanPahlevanan/Exercises/T1/ali-farhadian(alifrd49@gmail.com)/answer.pdf

1. NGINX Web Server (Static Site)

1. **Deploy an NGINX web server** to serve the HTML, CSS, and JavaScript files from the following repository: https://github.com/sahandghavidel/HTML-CSS-JavaScript-projects-for-beginners/tree/main/projects/age-calculator
2. **Summarize all relevant NGINX configuration files** used in this setup.
3. **Explain what an NGINX worker process is** and how it handles requests.
4. **Generate a self-signed TLS certificate** and configure HTTPS for this website.

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2. HAProxy vs NGINX

5. **Provide an overview of HAProxy** and explain how it differs from NGINX, including typical use cases for each.

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3. LXC Virtual Machines with NGINX

6. **Create two virtual machines using LXC** (or another virtualization engine such as VirtualBox or KVM).
7. **Install NGINX on both VMs** and configure each to serve a simple `index.html` file displaying:
    - VM1 → “vm1”
    - VM2 → “vm2”

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4. Load Balancing with Reverse Proxies

8. **Configure HAProxy as a reverse proxy** to load-balance traffic between VM1 and VM2.
9. **Configure NGINX as a reverse proxy** to load-balance traffic between VM1 and VM2.
10. **Implement rate limiting** on the reverse proxy and explain how it works.

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5. NFS Server Setup

11. **Set up an NFS environment using three virtual machines** (LXC, VirtualBox, or another VM engine):
    - VM1 → NFS server
    - VM2 and VM3 → NFS clients
12. **Configure VM2 and VM3 to mount and use the NFS share** hosted on VM1.

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6. Multi-Host NGINX (Name-Based Virtual Hosting)

13. **Create a multi-host NGINX setup** using two virtual machines:
    - VM1 serves `index.html` for domain `a.com`
    - VM2 serves `index.html` for domain `b.com`
14. **Configure NGINX on the host machine** so that:
    - `curl a.com` returns content from VM1
    - `curl b.com` returns content from VM2

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7. DNS Server Configuration (BIND)

15. **Set up a DNS server using BIND on VM1 or your local machine** with the following behavior:
    - Resolve `abc.com` to `1.2.3.4`
    - Forward all other DNS queries to:
        - Primary upstream DNS: `8.8.8.8`
        - Fallback DNS servers: `87.224.137.10`, `87.224.137.11`

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8. TLS Configuration

16. **Generate a self-signed TLS certificate** and configure HTTPS for **Exercise 1** using the domain `abc.com`.